



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 26-Feb-2025

Revision Number 0.1

## 1. Identification

### Product identifier

Product Name Liquichek Ethanol/Ammonia Control

### Other means of identification

Catalog Number(s) 544, 545, 546, 545X, 271, 272, 273, 272X

### Recommended use of the chemical and restrictions on use

Recommended use In vitro diagnostic

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories Inc.  
9500 Jeronimo Road  
Irvine, California 92618  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

#### Technical Service

1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Skin sensitization

Category 1

### Label elements

#### **Warning**

#### **Hazard statements**

May cause an allergic skin reaction



**Precautionary Statements - Prevention**

Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

IF ON SKIN: Wash with plenty of water and soap

If skin irritation or rash occurs: Get medical advice/attention

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Contains animal source material. (Cattle).

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name                                                                       | CAS No.    | Weight-%     | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|-------------------------------------------------------------------------------------|------------|--------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Ethyl alcohol                                                                       | 64-17-5    | 0.1 - 0.29   | -                                                                                     |                                                                   |
| Sodium fluoride                                                                     | 7681-49-4  | 0.1 - 0.249  | -                                                                                     |                                                                   |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.001 - 0.01 | -                                                                                     |                                                                   |
| Ammonium chloride                                                                   | 12125-02-9 | 0.001 - 0.01 | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                    |
| <b>Eye contact</b>    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician.                 |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation<br>or allergic reactions see a physician. |
| <b>Ingestion</b>      | Rinse mouth thoroughly with water.                                                                                                      |

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Itching. Rashes. Hives.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** May cause sensitization in susceptible persons. Treat symptomatically.

## **5. Fire-fighting measures**

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical** Product is or contains a sensitizer. May cause sensitization by skin contact.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## **6. Accidental release measures**

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## **7. Handling and storage**

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children.

## **8. Exposure controls/personal protection**

**Working area parameters, subject to mandatory control (MAC or TSEL)****Exposure Limits**

| Chemical name                   | Alberta                                                 | British Columbia                                        | Ontario                                                 | Quebec                                                  |
|---------------------------------|---------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| Ethyl alcohol<br>64-17-5        | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup>            | STEL: 1000 ppm                                          | STEL: 1000 ppm                                          | STEL: 1000 ppm                                          |
| Sodium fluoride<br>7681-49-4    | TWA: 2.5 mg/m <sup>3</sup>                              | TWA: 2.5 mg/m <sup>3</sup>                              | TWA: 2.5 mg/m <sup>3</sup>                              | TWA: 2.5 mg/m <sup>3</sup>                              |
| Ammonium chloride<br>12125-02-9 | TWA: 10 mg/m <sup>3</sup><br>STEL: 20 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>STEL: 20 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>STEL: 20 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>STEL: 20 mg/m <sup>3</sup> |

**Appropriate engineering controls**

**Engineering controls**                      Showers  
                                                          Eyewash stations  
                                                          Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection**                      Wear safety glasses with side shields (or goggles).

**Hand protection**                              Wear suitable gloves.

**Skin and body protection**                      Wear suitable protective clothing.

**Respiratory protection**                      No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations**                      Handle in accordance with good industrial hygiene and safety practice.

**9. Physical and chemical properties****Information on basic physical and chemical properties**

**Physical state**                                      Liquid  
**Appearance**                                      Clear to slightly cloudy  
**Color**                                                  colorless  
**Odor**                                                  Slight  
**Odor threshold**                                      No information available

| <b><u>Property</u></b>                         | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|------------------------------------------------|----------------------|--------------------------------|
| <b>pH</b>                                      | 6.95-7.05            |                                |
| <b>Melting point / freezing point</b>          | No data available    | None known                     |
| <b>Initial boiling point and boiling range</b> | No data available    | None known                     |
| <b>Flash point</b>                             | No data available    | None known                     |
| <b>Evaporation rate</b>                        | No data available    | None known                     |
| <b>Flammability</b>                            | No data available    | None known                     |
| <b>Flammability Limit in Air</b>               |                      | None known                     |
| <b>Upper flammability or explosive limits</b>  | No data available    |                                |
| <b>Lower flammability or explosive limits</b>  | No data available    |                                |
| <b>Vapor pressure</b>                          | No data available    | None known                     |
| <b>Relative vapor density</b>                  | No data available    | None known                     |
| <b>Relative density</b>                        | No data available    | None known                     |
| <b>Water solubility</b>                        | Miscible in water    |                                |
| <b>Solubility in other solvents</b>            | No data available    | None known                     |

|                           |                   |            |
|---------------------------|-------------------|------------|
| Partition coefficient     | No data available | None known |
| Autoignition temperature  | No data available |            |
| Decomposition temperature |                   | None known |
| Kinematic viscosity       | No data available | None known |
| Dynamic viscosity         | No data available | None known |

**Other information**

|                      |                 |
|----------------------|-----------------|
| Explosive properties | Not applicable. |
| Oxidizing properties | Not applicable. |
| Softening point      | Not applicable  |
| Molecular weight     | Not applicable  |
| VOC content          | Not applicable  |

**10. Stability and reactivity**

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

**11. Toxicological information****Information on likely routes of exposure****Product Information**

|              |                                                                                                                                                                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                                                                                                                                                        |
| Eye contact  | Specific test data for the substance or mixture is not available.                                                                                                                                                        |
| Skin contact | May cause sensitization by skin contact Specific test data for the substance or mixture is not available Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| Ingestion    | Specific test data for the substance or mixture is not available.                                                                                                                                                        |

**Symptoms related to the physical, chemical and toxicological characteristics**

|          |                         |
|----------|-------------------------|
| Symptoms | Itching. Rashes. Hives. |
|----------|-------------------------|

**Acute toxicity****Numerical measures of toxicity**

No information available

**Component Information**

| Chemical name            | Oral LD50            | Dermal LD50         | Inhalation LC50                                      |
|--------------------------|----------------------|---------------------|------------------------------------------------------|
| Ethyl alcohol<br>64-17-5 | = 7060 mg/kg ( Rat ) | -                   | = 116.9 mg/L ( Rat ) 4 h<br>= 133.8 mg/L ( Rat ) 4 h |
| Sodium fluoride          | = 52 mg/kg ( Rat )   | = 175 mg/kg ( Rat ) | -                                                    |

|                                                                                                |                      |                          |   |
|------------------------------------------------------------------------------------------------|----------------------|--------------------------|---|
| 7681-49-4                                                                                      |                      |                          |   |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat )   | = 87.12 mg/kg ( Rabbit ) | - |
| Ammonium chloride<br>12125-02-9                                                                | = 1650 mg/kg ( Rat ) | > 2000 mg/kg ( Rat )     | - |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.

**Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

**Respiratory or skin sensitization** May cause sensitization by skin contact.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

**Carcinogenicity** Based on available data, the classification criteria are not met.

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name                | ACGIH | IARC    | NTP   | OSHA |
|------------------------------|-------|---------|-------|------|
| Ethyl alcohol<br>64-17-5     | A3    | Group 1 | Known | X    |
| Sodium fluoride<br>7681-49-4 | -     | Group 3 | -     | X    |

**Legend**

**ACGIH (American Conference of Governmental Industrial Hygienists)**

A3 - Animal Carcinogen

**IARC (International Agency for Research on Cancer)**

Group 1 - Carcinogenic to Humans

Group 3 - Not Classifiable as to Carcinogenicity in Humans

**NTP (National Toxicology Program)**

Known - Known Carcinogen

**OSHA (Occupational Safety and Health Administration of the US Department of Labor)**

X - Present

**Reproductive toxicity** Based on available data, the classification criteria are not met.

**STOT - single exposure** Based on available data, the classification criteria are not met.

**STOT - repeated exposure** Based on available data, the classification criteria are not met.

**Aspiration hazard** Based on available data, the classification criteria are not met.

**12. Ecological information**

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

| Chemical name            | Algae/aquatic plants | Fish                                                                                                                                       | Toxicity to microorganisms | Crustacea                                                                        |
|--------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------|
| Ethyl alcohol<br>64-17-5 | -                    | LC50: 12.0 - 16.0mL/L (96h, Oncorhynchus mykiss)<br>LC50: >100mg/L (96h, Pimephales promelas)<br>LC50: 13400 - 15100mg/L (96h, Pimephales) | -                          | LC50: 9268 - 14221mg/L (48h, Daphnia magna)<br>EC50: =2mg/L (48h, Daphnia magna) |

|                                 |                                                                                                                    |                                                                                                                                                                                                 |   |                                                                                 |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------|
|                                 |                                                                                                                    | promelas)                                                                                                                                                                                       |   |                                                                                 |
| Sodium fluoride<br>7681-49-4    | EC50: =272mg/L (96h,<br>Pseudokirchneriella<br>subcapitata)<br>EC50: =850mg/L (72h,<br>Desmodesmus<br>subspicatus) | LC50: >530mg/L (96h,<br>Lepomis macrochirus)<br>LC50: =830mg/L (96h,<br>Lepomis macrochirus)<br>LC50: 38 - 68mg/L (96h,<br>Oncorhynchus mykiss)<br>LC50: =180mg/L (96h,<br>Pimephales promelas) | - | EC50: =338mg/L (48h,<br>Daphnia magna)<br>EC50: =98mg/L (48h,<br>Daphnia magna) |
| Ammonium chloride<br>12125-02-9 | -                                                                                                                  | LC50: =209mg/L (96h,<br>Cyprinus carpio)                                                                                                                                                        | - | -                                                                               |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

#### Component Information

| Chemical name                                                                                     | Partition coefficient |
|---------------------------------------------------------------------------------------------------|-----------------------|
| Ethyl alcohol<br>64-17-5                                                                          | -0.35                 |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.7                   |

**Other adverse effects** No information available.

### 13. Disposal considerations

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

**TDG** Not regulated

**DOT** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

### 15. Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

#### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

#### International Inventories

Contact supplier for inventory compliance status

## 16. Other information

|                                   |                         |                                  |                           |                                           |
|-----------------------------------|-------------------------|----------------------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b>                       | <b>Health hazards</b> 2 | <b>Flammability</b> 0            | <b>Instability</b> 0      | <b>Physical and chemical properties -</b> |
| <b>HMIS</b>                       | <b>Health hazards</b> 2 | <b>Flammability</b> 0            | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |
| <i>Chronic Hazard Star Legend</i> |                         | <i>* = Chronic Health Hazard</i> |                           |                                           |

#### Key or legend to abbreviations and acronyms used in the safety data sheet

##### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

#### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 RTECS (Registry of Toxic Effects of Chemical Substances)  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 26-Feb-2025

**Revision Note** Reviewed existing information and made minor updates.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

End of Safety Data Sheet